United International University

Department of Computer Science and Engineering

Object Oriented Programming Laboratory

Topic: Arrays (1D, 2D, Jagged) & Loops Duration: 2.5 hours

Lab Sheet: Arrays in Java

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1. Objectives

By the end of this lab, students will be able to:

- Declare, initialize, and use one-dimensional arrays.
- Work with two-dimensional (2D) arrays.
- Understand and implement jagged (ragged) arrays.
- Use different looping techniques: traditional `for` loop and enhanced `for-each` loop.
- Perform common array operations: sum, product, input, and output.
- Read array elements from user input using `Scanner`.

2. Introduction to Arrays in Java

An **array** is a collection of elements of the same data type stored in contiguous memory locations.

Key Concepts:

- Arrays are fixed in size once declared.
- Index starts from 0.
- **1D Array**: Single row of elements.
- **2D Array**: Matrix-like structure (rows and columns).
- **Jagged Array**: 2D array where each row can have a different number of columns.

3. Example 1: Array Initialization and Operations

ArrayInitialization.java

```java
public class ArrayInitialization {
    public static void main(String[] args) {
        int arr[] = {1, 2, 4, -5};

        // 1D Array Printing
        System.out.println("1D Array Elements:");
        for(int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();

        // Sum and Product of 1D Array
        int sum1D = 0, prod1D = 1;
        for(int i = 0; i < arr.length; i++) {
            sum1D += arr[i];
            prod1D *= arr[i];
        }
        System.out.println("Sum of 1D Array: " + sum1D);
        System.out.println("Product of 1D Array: " + prod1D);

        // 2D Array
        int arr2D[][] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};

        System.out.println("\n2D Array Elements:");
        for(int i = 0; i < arr2D.length; i++) {
            for(int j = 0; j < arr2D[i].length; j++) {
                System.out.print(arr2D[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

4. Example 2: Two-Dimensional Array with User Input

TwoDimensionalArrayInput.java

```java
import java.util.Scanner;

public class TwoDimensionalArrayInput {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int arr[][] = new int[2][2];

        System.out.println("Enter 4 elements for 2x2 matrix:");
        for(int i = 0; i < arr.length; i++) {
            for(int j = 0; j < arr[i].length; j++) {
                arr[i][j] = sc.nextInt();
            }
        }

        System.out.println("Matrix:");
        for(int i = 0; i < arr.length; i++) {
            for(int j = 0; j < arr[i].length; j++) {
                System.out.print(arr[i][j] + " ");
            }
            System.out.println();
        }
        sc.close();
    }
}
```

5. Example 3: Jagged Array

JaggedArrayDemo.java

```java
public class JaggedArrayDemo {
    public static void main(String[] args) {
        int two[][] = new int[3][];

        two[0] = new int[]{1, 2, 4, 6, 7};
        two[1] = new int[]{3, -1};
        two[2] = new int[]{9, 2, 6};

        int sum2 = 0;
        for (int i = 0; i < two.length; i++) {
            for(int j = 0; j < two[i].length; j++) {
                sum2 += two[i][j];
            }
        }
        System.out.println("Sum using traditional loop: " + sum2);
    }
}
```

6. Example 4: Enhanced For Loop

EnhancedForLoopDemo.java

```java
public class EnhancedForLoopDemo {
    public static void main(String[] args) {
        int two[][] = new int[3][];

        two[0] = new int[]{1, 2, 4, 6, 7};
        two[1] = new int[]{3, -1};
        two[2] = new int[]{9, 2, 6};

        int sum = 0;
        for(int i[] : two) {
            for (int j : i) {
                sum += j;
            }
        }
        System.out.println("Sum using enhanced for loop: " + sum);
    }
}
```

7. Tasks & Solutions

Task 1: 1D Array Operations (15 min)

(a) Create a program that takes 5 integers as input and stores them in a 1D array.

(b) Print the array, find the sum, and find the maximum element.

Task 2: 2D Matrix Input & Sum (20 min)

(a) Create a 3x3 matrix using user input.

(b) Print the matrix in proper format.

(c) Calculate and display the sum of all elements.

Solution Sketch:
Use nested loops for input and summation similar to the logic found in `TwoDimensionalArrayInput.java`.

Task 3: Jagged Array (20 min)

(a) Create a jagged array with 3 rows having 2, 4, and 3 columns respectively.

(b) Fill it with sample data and calculate the total sum using both traditional and enhanced for loops.

Task 4: Comprehensive Exercise (30 min)

Create a program that:

- Takes the size of a 1D array from the user and fills it dynamically.
- Creates a 2D jagged array based on user input.
- Displays all arrays and computes their sums securely.

Keep Coding!
Mastering arrays is essential for handling collections of data efficiently in Java.